

Unusual Does Not Mean Impossible

Unit 2 · Topic 2.10 · THE PROBLEM

Suppose the same 70 percent free-throw shooter takes 100 attempts under comparable conditions and makes 58. Let X be the number of makes if the career-rate model still applies.

Coach: “Anything under 60 makes is impossible for a 70 percent shooter, so 58 proves the career rate no longer applies.”

Analyst: “A result can be unusual without being impossible; compare 58 with the model’s mean and standard deviation.”

The binomial model gives a chance of about 0.0072 for 58 or fewer makes; use this provided value.

Career-rate model

Set	Attempts	Rate	Makes
Tonight	100	0.70	58

FIRST TAKE — one sentence before you work the parts

Whose statement is more defensible, the coach’s or the analyst’s? Name one quantity you would check.

- DOK 1** (a) State n and p for this model.
- DOK 2** (b) Calculate the binomial mean and standard deviation. Interpret the mean for many sets of 100 shots.
- DOK 3** (c) Whose statement is supported? Compare 58 with the mean and standard deviation, and use the given probability to distinguish unusual from impossible. Explain what the coach’s claim could make a reader wrongly believe. Write 3–4 sentences.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.